

CO3-AT3-ASSIGNMENT

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COURSE CODE: CSA0911

QUESTION-1:

CODE:

```
public class Main {  
    public static void main(String[] args) {  
        BankAccount account = new BankAccount();  
        DepositThread d = new DepositThread(account);  
        WithdrawThread w = new WithdrawThread(account);  
        d.start();  
        w.start();  
        try {d.join();  
            w.join();  
        } catch (InterruptedException e) {  
            System.out.println("Thread interrupted");}  
        System.out.println("All transactions completed.");  
    }  
}  
  
class BankAccount {  
    private int balance = 1000;  
    synchronized void deposit(int amount) {  
        balance += amount;  
        System.out.println("Deposited: " + amount);  
        System.out.println("Balance after deposit: " + balance);  
    }  
    synchronized void withdraw(int amount) {  
        if (amount <= balance) {  
            balance -= amount;  
            System.out.println("Withdrawn: " + amount);  
            System.out.println("Balance after withdrawal: " + balance);  
        } else {  

```

```

        System.out.println("Insufficient balance!");}}}}
class DepositThread extends Thread {
    BankAccount account;

    DepositThread(BankAccount account) {
        this.account = account;}

    public void run() {
        account.deposit(500);}}
class WithdrawThread extends Thread {
    BankAccount account;

    WithdrawThread(BankAccount account) {
        this.account = account;}

    public void run() {
        account.withdraw(300);}}

```

OUTPUT:

Your Code

```

1 public class Main {
2     public static void main(String[] args) {
3         BankAccount account = new BankAccount();
4         DepositThread d = new DepositThread(account);
5         WithdrawThread w = new WithdrawThread(account);
6         d.start();
7         w.start();
8         try {
9             d.join();
10            w.join();
11        } catch (InterruptedException e) {
12            System.out.println("Thread interrupted");
13        }
14        System.out.println("All transactions completed.");
15    }
16 }
17 class BankAccount {
18     private int balance = 1000;
19     synchronized void deposit(int amount) {
20         balance += amount;
21         System.out.println("Deposited: " + amount);
22         System.out.println("Balance after deposit: " + balance);
23     }
24     synchronized void withdraw(int amount) {
25         if (amount <= balance) {
26             balance -= amount;

```

Input

stdin input

Output

```

Deposited: 500
Balance after deposit: 1500
Withdrawn: 300
Balance after withdrawal: 1200
All transactions completed.

```

QUESTION-2:

CODE:

```
import java.util.Scanner;
import java.util.InputMismatchException;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String correctUsername = "admin";
        String correctPassword = "1234";
        try {
            System.out.print("Enter username: ");
            String username = sc.nextLine();
            System.out.print("Enter password: ");
            String password = sc.nextLine();
            if (!username.equals(correctUsername)) {
                throw new IllegalArgumentException("Invalid username");
            }
            if (!password.equals(correctPassword)) {
                throw new IllegalArgumentException("Invalid password");
            }
            System.out.println("Login successful!");
            System.out.print("Enter a number: ");
            int number = sc.nextInt();
            int result = 100 / number;
            System.out.println("100 / " + number + " = " + result);
        } catch (IllegalArgumentException e) {
            System.out.println("Login Error: " + e.getMessage());
        } catch (ArithmeticException e) {
            System.out.println("Arithmetic Error: Cannot divide by zero.");
        } catch (InputMismatchException e) {
            System.out.println("Input Error: Please enter a valid integer.");
        } finally {
```

```
System.out.println("Finally block executed.");  
sc.close();}}}
```

OUTPUT:

Your Code

```
1 import java.util.Scanner;  
2 import java.util.InputMismatchException;  
3 public class Main {  
4     public static void main(String[] args) {  
5         Scanner sc = new Scanner(System.in);  
6         String correctUsername = "admin";  
7         String correctPassword = "1234";  
8         try {  
9             System.out.print("Enter username: ");  
10            String username = sc.nextLine();  
11            System.out.print("Enter password: ");  
12            String password = sc.nextLine();  
13            if (!username.equals(correctUsername)) {  
14                throw new IllegalArgumentException("Invalid username");  
15            }  
16            if (!password.equals(correctPassword)) {  
17                throw new IllegalArgumentException("Invalid password");  
18            }  
19            System.out.println("Login successful!");  
20            System.out.print("Enter a number: ");  
21            int number = sc.nextInt();  
22            int result = 100 / number;  
23            System.out.println("100 / " + number + " = " + result);  
24        } catch (IllegalArgumentException e) {  
25            System.out.println("Login Error: " + e.getMessage());  
26        } catch (ArithmeticException e) {
```

Input

```
admin  
1234
```

Output

```
at  
java.base/java.util.Scanner.nextI  
nt(Scanner.java:2212)  
at  
main.main(main.java:21)
```

QUESTION-3:

CODE:

```
public class Main {  
    public static void main(String[] args) {  
        int[] numbers = {10, 20, 30, 40, 50};  
        SumThread sumThread = new SumThread(numbers);  
        AverageThread averageThread = new AverageThread(numbers);  
        MaximumThread maximumThread = new MaximumThread(numbers);  
        sumThread.start();  
        averageThread.start();  
        maximumThread.start();  
        try {  
            sumThread.join();  
            averageThread.join();  
            maximumThread.join();
```

```

        } catch (InterruptedException e) {
            System.out.println("Thread interrupted");}
        System.out.println("All calculations completed.");}}
class SumThread extends Thread {
    int[] numbers;
    SumThread(int[] numbers) {
        this.numbers = numbers;}
    public void run() {
        int sum = 0;
        for (int n : numbers) {
            sum += n;}
        System.out.println("Sum = " + sum);}}
class AverageThread extends Thread {
    int[] numbers;
    AverageThread(int[] numbers) {
        this.numbers = numbers;}
    public void run() {
        int sum = 0;
        for (int n : numbers) {
            sum += n;}
        double average = (double) sum / numbers.length;
        System.out.println("Average = " + average);}}
class MaximumThread extends Thread {
    int[] numbers;
    MaximumThread(int[] numbers) {
        this.numbers = numbers;}
    public void run() {
        int max = numbers[0];
        for (int n : numbers) {
            if (n > max) {
                max = n;}}
        System.out.println("Maximum = " + max);}}

```

OUTPUT:

Your Code

```
1 public class Main {
2     public static void main(String[] args) {
3         int[] numbers = {10, 20, 30, 40, 50};
4         SumThread sumThread = new SumThread(numbers);
5         AverageThread averageThread = new AverageThread(numbers);
6         MaximumThread maximumThread = new MaximumThread(numbers);
7         sumThread.start();
8         averageThread.start();
9         maximumThread.start();
10        try {
11            sumThread.join();
12            averageThread.join();
13            maximumThread.join();
14        } catch (InterruptedException e) {
15            System.out.println("Thread interrupted");
16        }
17        System.out.println("All calculations completed.");
18    }
19 }
20 class SumThread extends Thread {
21     int[] numbers;
22     SumThread(int[] numbers) {
23         this.numbers = numbers;
24     }
25     public void run() {
26         int sum = 0;
```

Input

stdin input

Output

```
Maximum = 50
Sum = 150
Average = 30.0
All calculations completed.
```

QUESTION-4:

CODE:

```
public class Main {
    public static void main(String[] args) {
        try {
            TrafficSignal red = new TrafficSignal("RED", 3000);
            TrafficSignal yellow = new TrafficSignal("YELLOW", 2000);
            TrafficSignal green = new TrafficSignal("GREEN", 3000);
            red.start();
            red.join();
            yellow.start();
            yellow.join();
            green.start();
            green.join();
            System.out.println("Traffic signal cycle completed.");
        } catch (InterruptedException e) {
```

```

        System.out.println("Main thread interrupted.");}}}}

class TrafficSignal extends Thread {

    private String signal;

    private int duration;

    TrafficSignal(String signal, int duration) {

        this.signal = signal;

        this.duration = duration;}

    public void run() {

        try {

            System.out.println(signal + " signal ON");

            Thread.sleep(duration);

            System.out.println(signal + " signal OFF");

        } catch (InterruptedException e) {

            System.out.println(signal + " thread interrupted"); }}}

```

OUTPUT:

Your Code

```

1 public class Main {
2     public static void main(String[] args) {
3         try {
4             TrafficSignal red = new TrafficSignal("RED", 1000);
5             TrafficSignal yellow = new TrafficSignal("YELLOW", 500);
6             TrafficSignal green = new TrafficSignal("GREEN", 1000);
7             red.start();
8             red.join();
9             yellow.start();
10            yellow.join();
11            green.start();
12            green.join();
13            System.out.println("Traffic signal cycle completed.");
14        } catch (InterruptedException e) {
15            System.out.println("Main thread interrupted.");
16        }
17    }
18 }
19 class TrafficSignal extends Thread {
20     private String signal;
21     private int duration;
22     TrafficSignal(String signal, int duration) {
23         this.signal = signal;
24         this.duration = duration;
25     }
26     public void run() {
27         try {

```

Input

stdin input

Output

```

RED signal ON
RED signal OFF
YELLOW signal ON
YELLOW signal OFF
GREEN signal ON
GREEN signal OFF

```